Preview: set-01/by_level/Grade_03-04_Ecolier/y2023_q15 (v1)

Fewest colours: find circles that are all joined to each other (a triangle needs 3), then colour step by step to show that many is enough

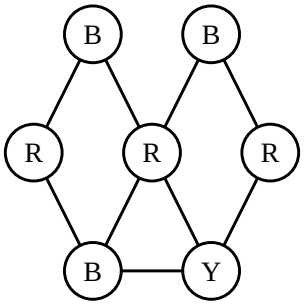
How to think about it: the fewest colours

Circles joined by a line must get different colours. To find the **fewest** colours, you need two things:

- **1. Why you need at least this many.** Look for circles that are **all** joined to each other. Three circles that form a triangle need 3 different colours.
- **2. Why this many is enough.** Really colour the picture, one circle after the other. Start with the circle that has the most lines. For each new circle, look at the circles it is joined to and pick a colour they do not have.

The contest question

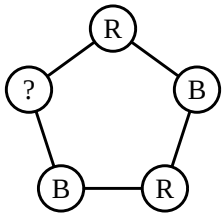
The middle circle and the two bottom circles form a triangle: you need at least 3 colours. And 3 colours are enough — here is one way (R = red, B = blue, Y = yellow):



Answer: **3** colours. Lines only care about the two circles at their ends — the left and the right circle may both be red, because no line joins them.

Watch out

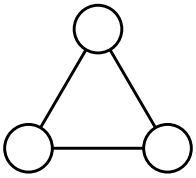
- No triangle does not always mean 2 colours! In a ring of 5 circles, red, blue, red, blue ... does not fit: the last circle touches red and blue. Count the circles in a ring: even is fine with 2, odd needs 3.



- Do not count the circles or the lines — many circles can share a colour.

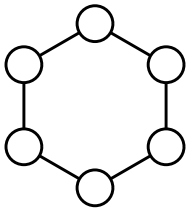
Warm-up: small steps first (no points)

W1. Circles joined by a line must get different colours. What is the smallest number of colours you need?



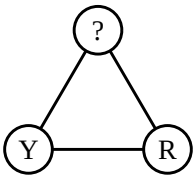
A. 3 ***B.*** 4 ***C.*** 5 ***D.*** 6 ***E.*** 7

W2. Circles joined by a line must get different colours. What is the smallest number of colours you need?



A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 6

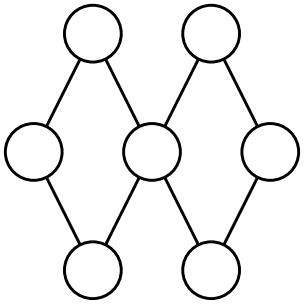
W3. Circles joined by a line must get different colours. You only have red (R), blue (B) and yellow (Y). Which colour must the circle with the question mark get?



A. blue ***B.*** any colour works ***C.*** red
D. it cannot be done ***E.*** yellow

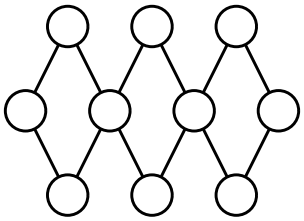
Part A: Each question is worth 3 points

1. Leo wants to colour in the circles in the diagram. When two circles are connected by a line they should have different colours. What is the minimum number of colours Leo needs?



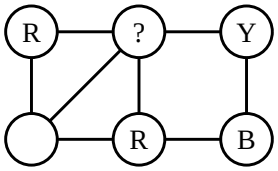
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 5 ***E.*** 8

2. Anna wants to colour in the circles in the diagram. When two circles are connected by a line they should have different colours. What is the minimum number of colours Anna needs?



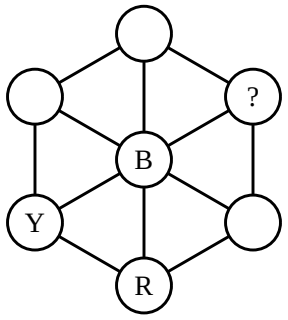
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 5 ***E.*** 12

3. Tom colours the circles red (R), blue (B) and yellow (Y). Circles joined by a line must get different colours. Some circles are already coloured. Tom wants to colour all the other circles too. Which colour must the circle with the question mark get?



A. it cannot be done ***B.*** blue ***C.*** red or blue, both work
D. yellow ***E.*** red

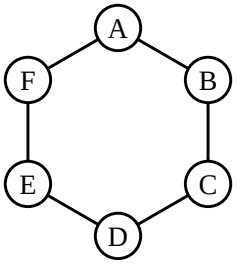
4. Sara colours the circles red (R), blue (B) and yellow (Y). Circles joined by a line must get different colours. Some circles are already coloured. Sara wants to colour all the other circles too. Which colour must the circle with the question mark get?



A. red ***B.*** it cannot be done ***C.*** blue
D. blue or yellow, both work ***E.*** yellow

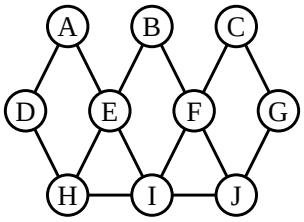
Part B: Each question is worth 4 points

5. A zoo has 6 animals, A to F. A line between two animals means that they do not get along, so they must not live in the same cage. Any number of animals can live in one cage. What is the smallest number of cages the zoo needs?



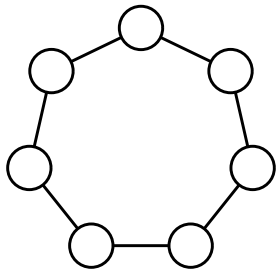
A. 1 cage ***B.*** 2 cages ***C.*** 3 cages ***D.*** 4 cages ***E.*** 6 cages

6. A zoo has 10 animals, A to J. A line between two animals means that they do not get along, so they must not live in the same cage. Any number of animals can live in one cage. What is the smallest number of cages the zoo needs?



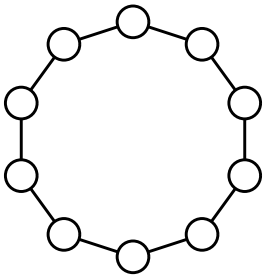
A. 1 cage ***B.*** 2 cages ***C.*** 3 cages ***D.*** 5 cages ***E.*** 14 cages

7. 7 children stand in a ring and hold hands, as in the picture. Two children who hold hands must wear T-shirts of different colours. What is the smallest number of colours needed?



A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 7

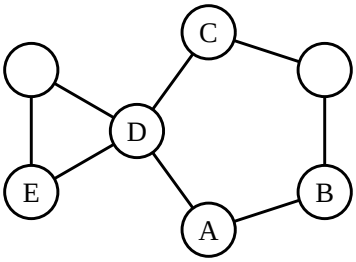
8. 10 children stand in a ring and hold hands, as in the picture. Two children who hold hands must wear T-shirts of different colours. What is the smallest number of colours needed?



A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 5 ***E.*** 10

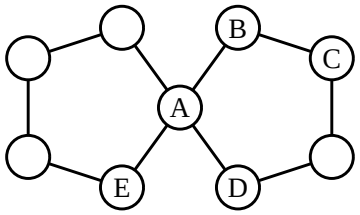
Part C: Each question is worth 5 points

9. Anna has only **two** colours. Circles joined by a line must get different colours. Anna may take away one circle together with its lines. Which circle must Anna take away, so that two colours are enough for the other circles?



A. circle A ***B.*** circle B ***C.*** circle C ***D.*** circle D ***E.*** circle E

10. Mia has only **two** colours. Circles joined by a line must get different colours. Mia may take away one circle together with its lines. Which circle must Mia take away, so that two colours are enough for the other circles?



A. circle A ***B.*** circle B ***C.*** circle C ***D.*** circle D ***E.*** circle E

Answer Key

W1.	**A**	**W2.**	**A**	**W3.**	**A**	**1.**	**B**	**2.**	**B**	**3.**	**B**
4.	**A**	**5.**	**B**	**6.**	**C**	**7.**	**C**	**8.**	**A**	**9.**	**D**
10.	**A**										

Solutions

W1. A (3) — Three circles that are all joined to each other need three different colours.
[set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · w_triangle]

W2. A (2) — Red, blue, red, blue, ... - opposite circles may share a colour, so 2 colours are enough.
[set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · w_square]

W3. A (blue) — It is joined to yellow and red, so it must be blue. [set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · w_next_colour]

1. B (2) — Two colours are enough: colour the circles in turn, so that every line joins two different colours.
[set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · fewest_colours]

2. B (2) — Two colours are enough: colour the circles in turn, so that every line joins two different colours.
[set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · fewest_colours]

3. B (blue) — The circle with the question mark must differ from the circles it is joined to. Only blue fits.
[set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · which_colour]

4. A (red) — The circle with the question mark must differ from the circles it is joined to, and those are forced by their own neighbours. Only red fits. [set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · which_colour]

5. B (2 cages) — A cage is like a colour: animals joined by a line need different cages. Share the animals out step by step: 2 cages are enough, and fewer do not work. [set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · zoo_cages]

6. C (3 cages) — A cage is like a colour: animals joined by a line need different cages. Look for animals that are all joined to each other. Then share the animals out step by step: 3 cages are enough, and fewer do not work.
[set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · zoo_cages]

7. C (3) — Red, blue, red, blue, ... works until the last child: with 7 children (an odd number) the last child holds hands with two different colours. So 3 colours are needed. [set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · hand_in_hand]

8. A (2) — Take turns: red, blue, red, blue, ... With 10 children (an even number) the ring closes nicely.
[set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · hand_in_hand]

9. D (circle D) — A ring with an odd number of circles cannot be coloured with two colours taking turns. There are two such rings (3 and 5 circles). Only circle D belongs to both rings, so taking it away breaks both of them.
[set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · take_one_away]

10. A (circle A) — A ring with an odd number of circles cannot be coloured with two colours taking turns. There are two such rings (5 and 5 circles). Only circle A belongs to both rings, so taking it away breaks both of them.
[set-01/by_level/Grade_03-04_Ecolier/2023 Q15 · take_one_away]